# The java.lang.Object Class

# The java.lang.Object Class

Every class extends the Java `Object` class.

Read that sentence again.

This class offers methods that every object is guaranteed to have.

**All Methods** | **Instance Methods** | **Concrete Methods**

| Modifier and Type | Method and Description |
|---|---|
| protected Object | **clone**()<br>Creates and returns a copy of this object. |
| boolean | **equals**(Object obj)<br>Indicates whether some other object is "equal to" this one. |
| protected void | **finalize**()<br>Called by the garbage collector on an object when garbage collection determines that there are no more references to the object. |
| Class<?> | **getClass**()<br>Returns the runtime class of this Object. |
| int | **hashCode**()<br>Returns a hash code value for the object. |
| void | **notify**()<br>Wakes up a single thread that is waiting on this object's monitor. |
| void | **notifyAll**()<br>Wakes up all threads that are waiting on this object's monitor. |
| String | **toString**()<br>Returns a string representation of the object. |
| void | **wait**()<br>Causes the current thread to wait until another thread invokes the **notify()** method or the **notifyAll()** method for this object. |
| void | **wait**(long timeout)<br>Causes the current thread to wait until either another thread invokes the **notify()** method or the **notifyAll()** method for this object, or a specified amount of time has elapsed. |
| void | **wait**(long timeout, int nanos)<br>Causes the current thread to wait until another thread invokes the **notify()** method or the **notifyAll()** method for this object, or some other thread interrupts the current thread, or a certain amount of real time has elapsed. |

# Equality

We often want to determine when two objects are the same.

But equality is a nebulous concept:



# Equality

- If two Person objects have the same first and last name are they the same?
  - What if they have different ages?
- When are two File objects the same?
  - If they have the same filename? The same content? At what point in time?
- When are two date objects the same?
  - If they refer to the same date? In what Calendar? What about leap seconds? Timezones?

# Equality in Java

We work with **two types of equality**:

**Reference Equality**

- Two objects are the same if they refer to the same physical memory location.
- We compare for reference equality with ==.
- Note that: `new Object() != new Object()`.

**Object Equality**

- Two objects are the same if the equals method **says so**.
- We compare for object equality with `o1.equals(o2)`.
- The default implementation is reference equality.
- The idea is that you override `equals` with your own implementation.
- If so, you must also override `hashCode`. If don't, disaster will follow.

# toString

The `toString` method returns a string representation of an object.

```
public static void main(String[] args) {
    Person person = new Person("Magnus");
    System.out.println(person);
}
```

```
Person@4554617c
```

```
@Override
public String toString() {
    return "Person{" + "name='" + name + '\'' + ", age=" + age + '}';
}
```

```
Person{name='Magnus', age=31}
```